SUPPLEMENTARY MATERIAL
CONSTELLATIONS FOR LYAPUNOV EXPONENTS

Each data file specifies nonnegative $(d+1)\times(d+1)$ matrices $A_0,\dots,A_{m-1}$, probabilities $\pi_0,\dots,\pi_{m-1}$, and an anchor $b\in\Delta^\circ$. For $x\in\mathbb{R}^{d+1}$, denote by $\|x\|_1$ the sum of the absolute values of its entries. Let $\Delta=\left\{x\in\mathbb{R}^{d+1}\,\middle|\,x_\ell\geq 0,\ \|x\|_1=1\right\}$, and denote by Δ° its relative interior. Let

$$f_i:\Delta^\circ\longrightarrow\Delta^\circ,\qquad f_i(x)=\frac{A_ix}{\|A_ix\|_1}.$$

Each file contains the support points x_j and weights α_j of a constellation

$$\eta=\sum_{j=1}^{N}\alpha_j\delta_{x_j},\qquad(\alpha_j\in\mathbb{R},\ x_j\in\Delta^\circ)$$

where $\sum_{j=1}^{N}\alpha_j=1$, and δ_x is the Dirac measure. The j-th bracketed word $[w_1,\dots,w_k]$ represents the point $x_j=f_{w_k}\cdots f_{w_1}(b)$. (The empty word $[\,]$ represents b.) It is then followed by its weight α_j, written as an exact rational number.

Let λ be the top Lyapunov exponent. Our approximation to λ is $\eta(\Phi)=\sum_{j=1}^{N}\alpha_j\Phi(x_j)$, where

$$\Phi:\Delta^\circ\longrightarrow\mathbb{R},\qquad\Phi(x)=\sum_{i=0}^{m-1}\pi_i\log\|A_ix\|_1\,.$$

From η, define

$$\sigma=\sum_{j=1}^{N}\alpha_j\left(\delta_{x_j}-\sum_{i=0}^{m-1}\pi_i\delta_{f_i(x_j)}\right)=\sum_{r=1}^{M}\beta_r\delta_{y_r}.$$

Let $a=b$, and define

$$K_a:\Delta^\circ\times\Delta^\circ\longrightarrow\mathbb{R},\qquad K_a(x,y)=\prod_{\ell=1}^{d+1}\frac{(x_\ell+a_\ell)(y_\ell+a_\ell)}{2a_\ell(x_\ell+y_\ell)}.$$

Clock inequality (Theorem 1.3 of the manuscript) gives the certified error bound

$$|\lambda-\eta(\Phi)|\leq\frac{\pi}{\sqrt{2}}\,\|\sigma\|_{K_a},\qquad\|\sigma\|_{K_a}^2=\sum_{1\leq r,s\leq M}\beta_r\beta_sK_a(y_r,y_s).$$

For Viswanath's constant (Example 2.1), $\rho=\exp\left((\sqrt{5}-1)\lambda/8\right)$ is the final value. For the random intersection of Sierpiński gaskets (Example 2.2), $\lambda/\log 2$ is its typical Hausdorff dimension. For the random intersection of Sierpiński carpets (Appendix E), $\lambda/\log 3$ is its typical Hausdorff dimension.